# Dian Ji

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## EDUCATION

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|---|---|
| 2023 – Present | **Ph.D.** in Earth, Environmental and Planetary Sciences GPA: 3.88 / 4.00<br>Department of Earth, Environmental and Planetary Sciences, Rice University<br>Advisor: Rajdeep Dasgupta, Committee: Cin-Ty Lee, Bidong Zhang |
| 2021 – 2023 | **M.Sc.** in Geology GPA: 4.00 / 4.00<br>Department of Earth and Planetary Sciences, University of Tennessee<br>Advisor: Nicholas Dygert, Committee: Molly McCanta, Shichun Huang, Bradley Thomson<br>Thesis: Numerical and Experimental Constraints on Trace Element Fractionation During Lunar Magma Ocean Solidification |
| 2016 – 2020 | **B.E.** in Resource Exploration Engineering<br>College of Geosciences, China University of Petroleum, Beijing<br>Advisor: Huichuan Liu |

*Google Scholar*

## PEER-REVIEWED PUBLICATIONS

**Total citations = 80; h-index = 5; i10-index = 3**

9. **D. Ji,** R. Dasgupta, (2026) Garnet pyroxenite at mid-mantle depths as a source of young lunar magmatism. *Earth and Planetary Science Letter.* doi.org/10.1016/j.epsl.2026.120133
8. C.T. Lee, J. Zhang, D. Keller, Y. Zhang, **D. Ji**, J. Mou, (2026) Extreme magmatic differentiation and silicic magmatism without low melt fractions. *Lithos.* doi.org/10.1016/j.lithos.2026.108581
7. **D. Ji,** R. Dasgupta, C.T. Lee, (2026) The effects of magmatic recharge on primary lunar melt compositions: Implications for the water and other volatile budget of the Moon. *Geochimica et Cosmochimica Acta.* doi.org/ 10.1016/j.gca.2026.03.042
6. Y. Zhang, R. Dasgupta, **D. Ji**, C. Lee, Y. Peng, B. Charlier, Z. Jin, J. Chen, O. Namur, (2025) Mantle melting conditions of mare lavas on South Pole–Aitken basin of lunar farside. *Geophysical Research Letters.* doi.org/10.1029/2024GL112418
5. **D. Ji,** R. Dasgupta, (2025) Sulfur inventory of the young lunar mantle constrained by experimental sulfide saturation of Chang'e-5 mare basalts and a new sulfur solubility model for silicate melts in equilibrium with sulfides of variable metal–sulfur ratio. *Geochimica et Cosmochimica Acta.* doi.org/10.1016/j.gca.2025.02.019
4. N. Dygert, **D. Ji,** E. Etheridge, (2025) A predictive model for divalent element partitioning between clinopyroxene and basaltic melt and a europium-in-plagioclase-clinopyroxene oxybarometer for cumulate rocks. *Geochimica et Cosmochimica Acta*. doi.org/10.1016/j.gca.2025.02.003
3. **D. Ji,** N. Dygert, (2024) Trace element partitioning between apatite and silicate melts: Effects of major element composition, temperature, and oxygen fugacity, and implications for the volatile element budget of the lunar magma ocean. *Geochimica et Cosmochimica Acta*. doi: 10.1016/j.gca.2023.11.004
2. **D. Ji,** N. Dygert, (2023) Trace element evidence for serial processing of the lunar flotation crust and a depleted bulk Moon. *Earth and Planetary Science Letters.* doi: 10.1016/j.epsl.2022.117958
1. **D. Ji**, H.C. Liu, Y.L. Li, (2020) Large-scale Early Cretaceous lower-crust melting derived adakitic rocks in NE China: implications for convergent bidirectional subduction and slab rollback. *International Geology Review*. doi:10.1080/00206814.2019.1697968

## FORTHCOMING

## CONFERENCE ABSTRACTS

15. **D. Ji,** R. Dasgupta, 2026. A garnet-clinopyroxenite mantle source for young lunar basalts. *2026 AGU Annual Meeting* submitted
14. **D. Ji,** R. Dasgupta, C. Lee, 2026. Was the lunar mantle ever water-enriched? New insights from magmatic recharge model. *Inner Solar System Workshop, Rice University*.
13. **D. Ji,** R. Dasgupta, C. Lee, 2025. Water-depleted mantle source for the water-enriched lunar magma suggested by the effects of magmatic recharge. *2025 AGU Annual Meeting* V21B-0065.
12. N. Dygert, E. Etheridge, R. Huk, **D. Ji,** 2025. Applications of a new Eu-in-plagioclase-clinopyroxene oxybarometer. *Goldschmidt Conference* #27264.
11. **D. Ji,** R. Dasgupta, 2025. The Existence of Garnet in the 2-Billion-Year-Old Lunar Mantle Indicated by High Pressure-Temperature Experiments and Trace Element Modeling. *Lunar and Planetary Science Conference, LVI* #1332.
10. N. Dygert, **D. Ji,** E. Etheridge, 2025. Toward a Clinopyroxene-Plagioclase Oxybarometer for Lunar and Terrestrial Cumulates: An $fO_2$-Dependent Predictive Model for Clinopyroxene-Melt Eu Partitioning. *Lunar and Planetary Science Conference, LV* #2419.
9. Y. Zhang, R. Dasgupta, **D. Ji**, C. Lee, Y. Peng, B. Charlier, Z. Jin, J. Chen, O. Namur, 2025. Mantle melting conditions in South Pole–Aitken basin. *Lunar and Planetary Science Conference, LVI* #2106.
8. **D. Ji,** R. Dasgupta, 2024. Deep Sulfur Cycle in the Young Lunar Mantle Constrained by High Pressure-Temperature Experiments on Sulfide Saturation of Chang'E 5 Mare Basalts. *2024 AGU Annual Meeting* P51E-3015.
7. **D. Ji,** N. Dygert, 2024. A New Europium in Apatite-Plagioclase Oxybarometer for Lunar and Terrestrial Cumulate Rocks and Meteorites. *Lunar and Planetary Science Conference, LV* #1240.
6. N. Dygert, **D. Ji,** E. Etheridge 2024. Toward a Clinopyroxene-Plagioclase Oxybarometer for Lunar and Terrestrial Cumulates: An $fO_2$-Dependent Predictive Model for Clinopyroxene-Melt Eu Partitioning. *Lunar and Planetary Science Conference, LV* #2419.
5. N. Dygert, **D. Ji**, 2023. Serial Processing of the Lunar Crust after the Magma Ocean Stage and a Depleted Bulk Moon: Insights from a Europium-in-Plagioclase Partitioning Model. *Goldschmidt Conference*, # 17023.
4. **D. Ji,** N. Dygert, 2023. New experimental constraints on REE partitioning between apatite and silicate melts and a temperature and composition-dependent predictive partitioning model. *Lunar and Planetary Science Conference, LIV* #1255.
3. **D. Ji,** N. Dygert, 2022. Serial processing after lunar anorthositic crust formation indicated by rare earth elements in plagioclase. *Lunar and Planetary Science Conference, LIII* #1229.
2. **D. Ji,** N. Dygert, 2021. Eu anomalies in lunar plagioclase reflect secondary processing by subsolidus reequilibration and introduction of a KREEP component. *Goldschmidt Conference*, #3219.
1. N. Dygert, **D. Ji**, A.L. Fagan, C.R. Neal, D.S. Draper, J.F. Rapp, T.J. Lapen, 2021. Petrogenesis of and subsolidus reequilibration within lunar ferroan anorthosites: Two demonstrations of a new $fO_2$-dependent model for plagioclase-melt europium partitioning. *Lunar and Planetary Science Conference*, LII, #2352.

**TALKS**

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| Jan 2026 | Graduate Interdisciplinary Earth Science Symposia (GIESS) Talk, Rice University |
| Nov 2025 | **Guest Lecturer** on *Trace Element Partitioning*, EEPS 322, Rice University |
| Mar 2025 | 56th Lunar and Planetary Science Conference, Houston |
| Feb 2025 | Graduate Interdisciplinary Earth Science Symposia (GIESS) Talk, Rice University |
| Feb 2025 | Lunch Bunch Talk in Geochemistry/Minerology/Petrology (GMP), Brown University **(Invited)** |
| Nov 2024 | **Guest Lecturer** on *Radiogenic Isotope Geochemistry*, EEPS 322, Rice University |
| Sep 2024 | Graduate Interdisciplinary Earth Science Symposia (GIESS) Talk, Rice University |
| Mar 2024 | 55th Lunar and Planetary Science Conference, Houston |
| Mar 2023 | 54th Lunar and Planetary Science Conference, Houston |
| Oct 2022 | **Guest Lecturer** on *Optical Mineralogy*, GEOL 310, University of Tennessee |
| Mar 2022 | 53rd Lunar and Planetary Science Conference, Houston |
| Jul 2021 | 31st Goldschmidt Conference, Virtual |

## GRANTS

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| 2023 – 2024 | **MSA Grant for Student Research in Mineralogy and Petrology**<br>Mineralogical Society of America<br>$5,000 to Student PI: Ji |

## HONORS & AWARDS

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| 2026 | **Alison Henning Teaching Award**, Rice University ($1,000) |
| 2025 | **AGU Annual Meeting Travel Award**, Rice University ($500) |
| 2025 | **Torkild Rieber Award in Geology**, Rice University ($2,500) |
| 2024 | **Planetary Origins to Habitability Graduate Fellowship**, Rice Space Institute (one semester stipend + $2,000 research funds) |
| 2024 | **AGU Annual Meeting Travel Award**, Rice University ($500) |
| 2023 – 2028 | **The Chair's Fellowship**, Rice University ($10,000) |
| 2023 | **Virginia & James Bibee Graduate Student Professional Promise Award**, University of Tennessee ($500) |
| 2023 | **Excellence in Teaching by GTA's Award**, University of Tennessee ($500) |
| 2023 | **Member**, *The Honor Society of Phi Kappa Phi* |
| 2023 | **LPSC Meeting Travel Award,** University of Tennessee ($500) |
| 2022 | **Jimmy Walls Colloquium Presentation Award**, University of Tennessee ($500) |
| 2022 | **LPSC Meeting Travel Award** University of Tennessee ($500) |
| 2020 | **Li Siguang Outstanding Student Award** (¥15,000) |
| 2020 | **Excellent Senior Thesis Award**, Beijing |
| 2020 | **Dean's Nomination Award**, China University of Petroleum (¥5,000) |
| 2019 | **First-class Scholarship**, China University of Petroleum (¥2,000) |
| 2018 | **Oriental Geophysics Company Scholarship** (¥3,000) |
| 2017 | **Second-class Scholarship**, China University of Petroleum (¥1,000) |

## OUTREACH & SERVICE

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| 2026 | Inner Solar System Workshop Session Chair, *Volatiles, Differentiation, and Evolution of Rocky Bodies* |
| 2026 | Reviewer for Graduate Women in Science (GWIS) National Fellowship |
| 2026 | Rice K-12 Earth and Planetary Open House |
| 2025 | Invited talk on planetary differentiation presented in the professional development series for K–12 science teachers in Houston. |
| 2024 | Volunteer Judge for Outstanding Student Poster Award, AGU Annual Meeting |

***Journal Reviewer***

Geochimica et Cosmochimica Acta × 3
Earth and Planetary Science Letter × 1
Icarus × 2
Lithos × 2
npj Space Exploration × 2
American Mineralogist × 1
International Geology Review × 2

## PROFESSIONAL TRAINING & EXPERIENCE

**Rice University**

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| Fall 2025 | EEPS 322: Chemistry of Earth and Planetary Materials, **Teaching Assistant** |
| 2022 – Summer 2025 | **Research Assistant** |

**University of Tennessee**

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| Spring 2023 | GEOL330: Igneous and Metamorphic Petrology, **Teaching Assistant**<br>Student evaluation 5.0 / 5.0 |
| Fall 2022 | GEOL310: Mineralogy, **Teaching Assistant** |

Student evaluation 4.8 / 5.0

2021 – 2022 **Research Assistant**

**University of Texas at Dallas**

Summer 2019 **Visiting scholar**
Supervisor: Robert J. Stern

## SUPERVISION

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| 2026 | **Naidhruv Ananth Iyer** (Undergraduate, Rice University)<br>Trace element partitioning experiments |
| Summer 2025 | **Denim-Leenox Sasser** (High school intern)<br>Garnet stability in the lunar mantle |
| Summer 2024 | **Aahan Roy** (High school intern)<br>Sulfur solubility of lunar basalts<br>→Now a undergraduate student at UT Austin |
| 2022 – 2023 | **Jordan Marshall** (Undergraduate, University of Tennessee)<br>Piston-cylinder experiments<br>→ Materials Engineering, University of Tennessee (PhD track) |

## FIELD EXPERIENCE

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| 2025, 2023 | **General field trip, New Mexico**<br>A field trip of igneous and metamorphic rocks in New Mexico for a week led by Dr. Cin-Ty Lee and Dr. Rajdeep Dasgupta |
| 2022 | **Rio Grande Rift and Jemez Lineament xenolith sampling, New Mexico**<br>Collected mantle and crustal xenolith from Kilbourne Hole to Cerro de Guadalupe in New Mexico for a week, led by Dr. Nicholas Dygert |
| 2022 | **McClung Blue Ridge Foothills Field Trip**<br>Observed part of the transition from the external foreland fold-thrust belt of the Appalachians into the internal metamorphic core, led by Dr. Bob Hatcher |
| 2019 | **Archean Basic Rock Collection, Miyun**<br>Collected Archean garnet pyroxenite and measured geological occurrence of basaltic dyke group, led by Dr. Huichuan Liu |
| 2019 | **Field Practice in Oilfield, Dagang Oilfield**<br>Learned the working methods of oilfield engineers, and interpretation of seismic data as well as logging data for two weeks |
| 2018 | **Comprehensive Geological Field Practice, Liujiang Basin**<br>A month-long geological field practice includes surveys of stratigraphic profiles and geological mapping, led by Dr. Liang Luo |
| 2017 | **General Field Practice, Western Hills of Beijing**<br>A two-week geological field practice for learning magmatic, sedimentary, and metamorphic rocks, led by Dr. Qin Zhang |

Last updated: Aug 8, 2026